

Mahi Kanchan and RCB 911 are hybrid varieties of Maize and Bajara respectively. Significantly the major maize producing districts in the state of Rajasthan are Bhilwara, Chhittorgarh, Udaipur, Banswara etc. While Western Rajasthan, Jaipur, Dausa, Karauli, Bharatpur etc. are major Bajra producing regions of Rajasthan.

105. The art and science of cultivating forest crops is known as:

- (a) Silviculture
- (b) Agroforestry
- (c) Arboriculture
- (d) Forestry

Chhattisgarh P.C.S. (Pre) 2024

Ans. (a)

Silviculture is the art and science of controlling the establishment, growth, composition, health and quality of forests and woodlands to meet the diverse needs and values of landowners and society such as wildlife habitat, timber, water resources, restoration and recreation on a sustainable basis.

Animal Husbandry

*Animal Husbandry refers to the management and care of animals for purposes such as obtaining food, fiber, labor and other products. It includes practices to improve the quality and productivity of livestock through breeding, feeding and disease control. As per 20th Livestock census total cattle population in the country is 193.46 million as of 2019. In India, there are mainly two types of cattle breeds.

(i) **Milch Breeds** - The cow of this breed give more milk but the Bull are not good for agricultural work and load vehicle. It includes major breeds like **Gir, Sahiwal, Sindhi, Devni, Tharpakar**, etc.

(ii) **Drought Breeds** - It includes breeds like **Nagauri, Bhalavi, Hallikar, Kangayam, Killari, Powar Seri** etc. The country's best indigenous cow breeds are found in Rajasthan. The cows breed found in Rajasthan are **Tharparkar, Rathi, Kakarange, Sahiwal, Lal Sindhi, Gir, Nagauri, Haryanvi, Malvi** and **Mewati**. *Sahiwal cow breeds is mainly found in Sindh province of Pakistan. Its milking capacity is quite high.

There are many breeds of buffalo. They mainly include Murra, Bhadawari, Zafara badi, Suttari, Mahsana, Nagpuri, Neeli, Pandharpuri etc.

*According to Basic Animal Husbandry Statistics 2024, in India, maximum sheep are found in Telangana in 2019. The best quality of sheep breed of the country are found in the valley of Kashmir, Kullu, Chamba and Kangada.

*Goat are known as the "**cow of poor**". Milk, meat, leather and wool are obtained from the goats. In India, they are the main source of meat. The highest milk yielding breed of goat in India is **Jamunapari**. It is also considered as good for both milk and meat. It producer about 2.5-3.0 kg milk everyday.

*According to data of Basic Animal, Husbandry Statistics 2024, the total meat production in the country is 10.25 million tonnes during 2023-24. The meat production has registered growth rate 4.95% in the Year 2023-24. The top 5 meat producing states are West Bengal (12.62%), Uttar Pradesh (12.29%), Maharashtra (11.28%), Telangana (10.85%) and Andhra Pradesh (10.42%).

In India, the white revolution was started in July, 1970 with the launch of operation flood-I Plan. **Dr. Verghese Kurien** is known as the **father of the white revolution** in India. White revolution is related to milk production. According to production data of Food and Agriculture Organization Corporate Statistical Database (FAOSTAT), India is the highest milk (Raw milk of cattle) producer i.e., rank first position in the world contributing 21.78% of global milk production in the year 2023. According to Basic Animal Husbandry Statistics 2024, total milk production in the country is 239.30 million tonnes during 2023-24. The milk production has increased by 3.78% over the previous year (2022-23). The per-capita availability of milk is 471 grams per day. The average yield per animal and for indigenous/non descript is 3.54 kg/day/Animal per day for exotic/crossbred is 8.43 kg/day/Animal and for indigenous/non-descript is 3.54 kg/day/Animal. Top 5 milk producing states are Uttar Pradesh (16.21%), Rajasthan (14.51%), Madhya Pradesh (8.91%), Gujarat (7.65%), Maharashtra (6.71%). Operation Flood was started to increase the production of milk. Its first phase was from 1970 to 80, the second phase from 1981 to 85 and the third phase from 1985 to 1996. The total egg production in India during 2023-24 is 142.77 billion nos. The top 5 egg-producing states are Andhra Pradesh (17.85%), Tamil Nadu (15.64%), Telangana (12.88%), West Bengal

(11.37%) and Karnataka (6.63%).

*In the year 1970, **Indian Dairy corporation** was established in **Anand** (Gujarat). Its aim was to make Operation Flood successful. National Dairy Board was established in 1965.

Dr. Verghese Kurien was the founder of the National Dairy Development Board and the Indian Dairy Corporation.

National Dairy Research Institute is located in Karnal, Haryana. It was given the status of Deemed University in 1989. **Bihar** has the highest animal density in India. In the context of total livestock in 2019, Uttar Pradesh was at the top and Rajasthan and Madhya Pradesh were placed at second and third respectively. In the context of cattle stock, the top three states are West Bengal, Uttar Pradesh and Madhya Pradesh.

1. Stock Farming is –

- (a) Growing of 2-3 crops at the same time
- (b) Breeding of animals
- (c) Crop rotation
- (d) None of the above

U.P.P.C.S. (Mains) 2005

Ans. (b)

Stock farming is the practice of keeping livestock farm. Animal Husbandry is the scientific management of animal livestock. It includes various aspect such as feeding, breeding and disease control. Thus (b) is the correct answer.

2. Largest density of the number of cattle in per 100 hectare gross cultivable land is in –

- (a) Bihar
- (b) Haryana
- (c) Madhya Pradesh
- (d) Uttar Pradesh

I.A.S. (Pre) 1994

Ans. (a)

States and density of number of cattle in their gross cultivable land are given in the following order

Haryana	-	25/square km.
Uttar Pradesh	-	77/square km.
Madhya Pradesh	-	106/square km.
Bihar	-	183/square km.

3. About one-third of India's cattle population is found in the three states, these are :

- (a) Bihar, Maharashtra, and Uttar Pradesh
- (b) Madhya Pradesh, West Bengal, and Uttar Pradesh

(c) Punjab, Orissa, and Rajasthan

(d) Andhra Pradesh, Karnataka, and Rajasthan

U.P.P.C.S. (Pre) 2009

U.P.P.C.S. (Mains) 2006

Ans. (*)

When the question was asked option (b) was correct answer. According to 20th Livestock census, there are 193.46 million cattle in India. Top three states with cattle populations (in 2019) are as follows - West Bengal (19.1 million), Uttar Pradesh (19.0 million) and Madhya Pradesh (18.8 million).

4. Which of the following statements are correct ?

Select the correct answer using the code given below-

- 1. The largest number of cattle are found in Madhya Pradesh
- 2. The largest number of Buffaloes in India are found in Uttar Pradesh
- 3. The largest number of sheep in India are found in Rajasthan
- 4. Uttar Pradesh in India is the largest producer of cow milk.

(a) 1 and 2

(b) 2 and 3

(c) 3 and 4

(d) 1 and 4

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (*)

When the question was asked option (a) was correct answer. According to the 20th Livestock Census, 2019, three states with highest livestock population are - Uttar Pradesh, Rajasthan and Madhya Pradesh.

Top 3 buffaloes owner states are Uttar Pradesh, Rajasthan and Gujarat. Top 3 Sheep owner states – Telangana > Andhra Pradesh > Karnataka. According to Basic Animal Husbandry Statistics 2024, Uttar Pradesh in India is the largest producer of milk in 2023-24.

5. Most milk yielding goat breed in India is-

- (a) Barbari
- (b) Jamunapari
- (c) Kali Bengali
- (d) Beetal

I.A.S. (Pre) 1994

Ans. (b)

Most milk yielding goat breed in India is Jamunapari goat. This breed is considered useful for milk as well as for meat. It gives up to about 2.5 to 3.0 kg of milk, per day. In 2023-24, the indigenous cattle contribute 11.36% of the total milk production in the country whereas non-descript cattle contribute 10.11% and non-descript buffaloes contribute 13.83% of the total milk production in the country. Goat milk share a contribute of 3.36% in the total milk production across the country.

6. Where are the Tharparkar species found?

- (a) Tribal Belt
- (b) Border Region of Rajasthan
- (c) Hadauti Region
- (d) Torawati Region

R.A.S./R.T.S.(Pre) 2007

Ans. (b)

Best cow breeds in India are found in Rajasthan. Breeds of cow generally found in Rajasthan are Tharparkar, Rathi, Kankrej, Sahiwal, Lal Sindhi, Gir, Nagouri, Haryanvi, Malwi and Mewati. Among these, Tharparkar breed is found in the western border district of Jaisalmer, Barmer and Jodhpur. Jaisalmer district of Rajasthan is at the top in terms of per-capita milk production. Apart from this, it is also found in Kutch area of Gujarat.

7. Which of the following is not a breed of Rajasthan?

- (a) Tharparkar
 - (b) Rathi
 - (c) Hallikar
 - (d) Mewati
- (e) None of the above / More than one of the above

66th B.P.C.S. (Pre) (Re-Exam) 2020

Ans. (c)

Rajasthan state has three native cattle breeds viz Rathi, Tharparkar and Nagori, having great deal of endurance. Hallikar is a breed of cattle native to the state of Karnataka. Mewati, also known as kesi, is a breed of Haryana.

8. The cow breed known for high milk yield is

- (a) Gangotri
- (b) Haryana
- (c) Sahiwal
- (d) Tharparkar

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (c)

Sahiwal breed of cow is mainly found in Sindh-province of Pakistan. Its milk yielding capacity is very high. Gangotri is mainly found in Uttar-Pradesh and its milk yielding capacity is very low. Haryana and Tharparkar are mainly used for dual purpose i.e., as a carriers and for milk yielding.

9. India's place in milk production in the world is-

- (a) First
- (b) Second
- (c) Third
- (d) Fourth

U.P.P.C.S. (Spl.) (Pre) 2008

Ans. (a)

India is ranked first in terms of milk production. In India, Uttar Pradesh was at first place, and Rajasthan in the second place during 2023-24.

10. Which one of the following states recorded the highest milk production in 2011-12 ?

- (a) Uttar Pradesh
- (b) Rajasthan
- (c) Gujarat
- (d) Punjab

U.P.P.C.S. (Pre) 2014

Ans. (a)

States given in the options and their milk production during 2011-12 and 2023-24 is given below –

States	Milk Production (thousand tonnes)	
	(2011-12)	(2023-24)
Uttar Pradesh	22556	38779.57
Rajasthan	13512	34733.00
Gujarat	9817	18311.80
Punjab	9551	14000.18

11. What is 'Operation Flood' associated with?

- (a) Flood Control
- (b) Irrigation Project
- (c) Storage of Foodgrains
- (d) Milk Production and Distribution

M.P.P.C.S. (Pre) 1998

Ans. (d)

Operation Flood was a revolutionary initiative launched in 1970 by the National Dairy Development Board (NDDB) of India. Spearheaded by Dr. Verghese Kurien often called the "Father of the White Revolution", this programme transformed India from a milk deficient country to the largest producer of milk in the world.

12. Who is known as the father of India's 'White Revolution'?

- (a) Dr. Varghese Kurien
- (b) Dr. Hargovind Khurana
- (c) Dr. Homi Bhabha
- (d) Dr. Rajendra Prasad

Uttarakhand P.C.S. (Mains) 2006

Ans. (a)

Dr. Varghese Kurien was the Father of India's White Revolution, which helped India emerge as the largest milk producer in the world.

13. White Revolution is related to –

- (a) Egg production
- (b) Milk production
- (c) Wheat production
- (d) Fish production

U.P.P.C.S. (Pre) 2015

Ans. (b)

See the explanation of above question.

14. Who among the following is associated with 'White Revolution'?

- (a) P.J. Kurien
- (b) Varghese Kurien